

Name: _____ **Date:** _____

Problem 1. For each of the following equations, use implicit differentiation to solve for $\frac{dy}{dx}$

a) $x^2 + y^2 = 1$ (a circle centered at the origin)

b) $\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$ (an ellipse centered at the origin)

c) $x^2 + xy + y^2 = 7$

d) $(x + y)^{\frac{2}{3}} = y$

e) $x = 1 + y - y^3$

Problem 2. For each of the following equations and points, find the equation of the tangent line to the graph at the given point (you do not need to recalculate the derivatives, you can use your answers from the previous problems!)

a) $x = 1 + y - y^3$ at $(1, 1)$

b) $x^2 + xy + y^2 = 7$, at $(1, 2)$

c) $x^2 + y^2 = 1$, at $(\frac{1}{2}, \frac{\sqrt{3}}{2})$

d) $(x + y)^{\frac{2}{3}} = y$, at $(0, 1)$

e) $x = 1 + y - y^3$, at $(1, 0)$